# 22秋线代期末试卷(回忆版)

1. Multiple Choice. Only one choice is correct.

(1) Let $A, B, C$ be $n \times n$ matrices with $B$ invertible and $AB = C$. Which of the following must be true?

(A) The row spaces of $A$ and $C$ are the same.

(B) The null spaces of $A$ and $C$ are the same.

(C) The column spaces of $A$ and $C$ are the same.

(D) The determinants of $A$ and $C$ are the same.

(2) Let $P$ be a $5 \times 5$ permutation matrix. Which of the following is false?

(A) $P$ is an orthogonal matrix.

(B) $P$ must have real eigenvectors.

(C) There always exists an invertible real matrix $Q$ such that $Q^{-1}PQ$ is diagonal.

(D) The equation $Px = 0$ has only zero solution.

(3) Let $A$ be an $n \times n$ real symmetric matrix. Which of the following statements must be true?

(A) $A$ must have $n$ distinct eigenvalues.

(B) Some of the complex eigenvalues of $A$ need not be real.

(C) Any $n$ linearly independent ~~eigenvalues~~ eigenvectors of $A$ are pairwise orthogonal.

(D) There is an orthogonal matrix $Q$, such that $Q^TAQ$ is diagonal.

(4) Let $\alpha_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$, $\beta_1 = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}$, $\beta_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

If $\gamma$ can be written as a linear combination of $\alpha_1, \alpha_2$, and $\gamma$ can also be written as a linear combination of $\beta_1, \beta_2$, then $\gamma$ has the form

(A) $k\begin{bmatrix} 1 \\ 5 \\ 8 \end{bmatrix}, k \in R$. (B) $k\begin{bmatrix} 3 \\ 5 \\ 10 \end{bmatrix}, k \in R$. (C) $k\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, k \in R$. (D) $k\begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}, k \in R$.

(5) Which of the following matrices is congruent to the identity matrix?

(A) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. (B) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 7 & 1 \\ 1 & 1 & 8 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & -1 & 2 \\ -1 & 3 & -3 \\ 2 & -3 & -4 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.

2. Fill in the blanks.

(1) Let $A$ be a $2 \times 2$ matrix, which has two linearly independent eigenvectors $v_1$ and $v_2$ such that $A^2(v_1 - v_2) = 2v_1 + v_2$. Then $\det(A^4) =$ _____.

(2) The singular values of the matrix $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$ are _____.

(3) Let $A$ be a $3\times3$ matrix which has eigenvalues $-1,0,1$. Suppose that $(A+aI_3)A(A-bI_3)=0$, where $I_3$ is the $3\times3$ identity matrix. Then $a=$_____, $b=$_____.

(4) If $A=\begin{bmatrix}0&0&1\\x&1&2x-3\\1&0&0\end{bmatrix}$ is diagonalizable, then $x=$_____.

(5) Let $A$ be a $4\times4$ symmetric matrix such that $A^2+A=0$. Suppose that $A$ has rank 3. A diagonal matrix that is similar to $A$ is _____.

3. Let $A_n$ be the $n\times n$ matrix

$$A_n=\begin{bmatrix}1&-a&0&0&\cdots&0\\-a&1&-a&0&\cdots&0\\0&-a&1&-a&\cdots&0\\\vdots&&\ddots&\ddots&\ddots&\vdots\\0&0&\cdots&-a&1&-a\\0&0&\cdots&0&-a&1\end{bmatrix}$$

(a) Find constants $b,c$ such that the sequence $\det(A_n)$ satisfies

$$\det(A_n)=b\cdot\det(A_{n-1})+c\cdot\det(A_{n-2}) \quad \text{for all } n\geq3.$$

(b) Find a matrix $B$ such that $x_n=Bx_{n-1}$ for $n\geq3$, where $x_n=\begin{bmatrix}\det(A_n)\\\det(A_{n-1})\end{bmatrix}$.

(c) For $a^2=\frac{3}{16}$, find an expression for $\det(A_n)$ for all $n\geq3$.

4. Suppose $\alpha,\theta\in(0,\frac{\pi}{2})$.

(a) Compute $A_n=\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}^n\begin{bmatrix}9&0\\0&4\end{bmatrix}\begin{bmatrix}\cos\alpha&\sin\alpha\\-\sin\alpha&\cos\alpha\end{bmatrix}^n$ for all $n\geq1$.

(b) Find a singular value decomposition (SVD) of $A_n$ for each $n\geq1$.

(c) Show that the matrix $A_1$ is symmetric if and only if $\alpha=\theta$.

(Hint: the formula $\sin(\theta-\alpha)=\sin\theta\cos\alpha-\cos\theta\sin\alpha$ may be useful.)

(d) Prove that if $A_1$ is symmetric, then $A_n$ is positive definite for every $n\geq1$.

5. Consider the quadratic form $f(x_1,x_2,x_3)=x_1^2+2x_2^2-2x_3^2-4x_1x_3$.

(a) Find the symmetric matrix $A$ such that $f(x)=x^TAx$ for all $x=(x_1,x_2,x_3)^T$, and find an orthogonal matrix $Q$ such that $Q^TAQ$ is a diagonal matrix.

(b) The quadric surface defined by the equation $f(x,y,z)=2023$ is _____.

(A) a hyperboloid of one sheet (B) a hyperboloid of two sheets

(C) an ellipsoid (D) none of the above

6. For any $a=(a_1,\cdots,a_n)^T\in R^n$, put $\|a\|=\sqrt{a_1^2+\cdots+a_n^2}$. Let $x,y\in R^n$ be nonzero vectors.

(a) Show that if there is an orthogonal matrix $S$ such that $Sx=y$, then $\|x\|=\|y\|$.

(b) Let $N$ be the null space $N(x^T)$ of the $1\times n$ matrix $x^T$. Show that $\dim N=n-1$.

(c) Let $\alpha_2,\cdots,\alpha_n$ be a basis of $N$. Show that the system $\alpha_1:=x,\alpha_2,\cdots\alpha_n$ is linearly ~~id~~independe independent.

(d) Let $A$ be the matrix with $\alpha_1, \alpha_2, \cdots, \alpha_n$ as its columns. Let $A = QR$ be a factorization with $Q$ orthogonal and $R$ upper triangular. Write $R = (r_{ij})$. Show that $|r_{11}| = \|x\|$.

(e) Prove that if $\|x\| = \|y\|$, then there exists an orthogonal matrix $S$ such that $Sx = y$.